

Case Report

Utilizing a Deep Eutectic Solvent in carotene extraction technology with Palm Methyl Ester oil to advance green technology

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ABSTRACT

In this research aims to develop and evaluate the use of Deep Eutectic Solvent (DES) in carotene extraction technology with Palm Methyl Ester (PME) as a green solvent to support the development of environmentally friendly technology. The carotenoids existing in palm biodiesel will be separated via a liquid-liquid extraction technique that employs methanol as the solvent along with deep eutectic solvent (DES) serving as a co-solvent. The DES utilized comprises betaine serving as a hydrogen bond donor (HBD) in conjunction with citric acid and glycerol acting as hydrogen bond acceptors (HBA), reacting in a molar ratio of 1:1 to 1:6. The produced deep eutectic solvent (DES) is characterized via Fourier Transform Infrared Spectroscopy (FTIR) for the purpose of identifying the functional groups created, while Proton Nuclear Magnetic Resonance (^1H NMR) is employed to delineate the structural configuration of the components formed within the DES. After the DES solution is prepared, its key physical properties, including pH, density and viscosity, will be measured and analyzed. The total carotenoids extracted are examined utilizing UV-Visible Spectrophotometry and High-Performance Liquid Chromatography (HPLC) to identify the concentration of the beta-carotene component acquired. The optimal extraction condition for total carotenoids is established at 2500 ppm utilizing DES betaine: acetic acid with an Ester: DES ratio of 1:2, whereas the total carotenoid concentration obtained with DES betaine: glycerol at an Ester: DES ratio of 1:4 is 2287 ppm. The beta-carotene concentration attained with DES Betaine: acetic acid is quantified at 33.63 IU/g, whereas with DES betaine: glycerol, it is recorded at 22.58 IU/g.

1. Introduction

In the preceding decade, advancements in technology pertaining to green chemistry have witnessed significant acceleration. A notable innovation in this realm is the establishment of extraction methodologies employing environmentally benign solvents. The technology for environmentally conscious extraction is primarily facilitated by two principal solvents, specifically Ionic Liquid (IL) and Deep Eutectic Solvent (DES) [1]. IL represents the inaugural environmentally sustainable solvent devised as a substitute for traditional organic solvents. The application of IL as a solvent in the extraction and/or purification of bioactive constituents demonstrates efficacy for both simple organic entities and more intricate molecular structures. The utilization of IL has been shown to yield superior extraction efficiencies and purification

ratios in comparison to conventional materials and organic solvents [2]. Nevertheless, contemporary investigations into IL have revealed its significant volatility, flammability, instability, and potential toxicity. Furthermore, the synthesis of IL typically incurs substantial costs and complexity, as it cannot be generated in a singular reaction step and requires an extended duration for production [3]. In light of these considerations, Deep Eutectic Solvent (DES) has been formulated to address the limitations associated with IL. DES comprises a mixture of chemical constituents that serve as hydrogen bond acceptors (HBA) and hydrogen bond donors (HBD). These chemical constituents are characterized by their non-toxic and biodegradable nature. The conceptual framework of DES was initially proposed by Abbott through his investigative work on the amalgamation of quaternary ammonium salts and urea [4]. In comparison to IL, the synthesis of DES can be executed

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through more streamlined and cost-effective methodologies. DES finds application across a spectrum of extraction processes, encompassing metal recovery in mining as well as the extraction of bioactive substances from natural sources [5].

Palm oil represents a category of vegetable oil characterized by its abundant phytonutrient content, including tocopherols, tocotrienols, carotenoids, sterols, squalene, phospholipids, and polyphenols. The unique reddish-orange hue of palm oil can be attributed to the presence of various carotenoids [6]. Crude palm oil (CPO) comprises carotenoid concentrations ranging from 500 to 700 ppm, predominantly consisting of α -carotene and β -carotene, which together account for approximately 90 % of the total carotenoid content in CPO. Both α and β -carotene exhibit provitamin A activity, with respective activity values quantified at 0.9 and 1.67. Furthermore, carotenoids are endowed with antioxidant properties, which may contribute to a reduction in cancer risk and possess anti-atherosclerotic effects. The nutritional advantages and chromatic properties of carotenoids render them valuable as natural food colorants and in the formulation of pharmaceutical products [7].

Methyl ester constitutes an environmentally sustainable, renewable, and non-toxic fuel synthesized from oils or fats via a transesterification process utilizing alcohol. Crude biodiesel, originating from palm oil, encompasses by-products referred to as minor components, which include free fatty acids (FFA), monoglycerides (MG), diglycerides (DG), triglycerides (TG), alongside antioxidant substances such as carotenoids (α - and β -carotene) and vitamin E (tocopherols and tocotrienols). The minor components present in crude palm oil are characterized by carotenoid concentrations of 500–700 ppm, with over 80 % attributed to α - and β -carotene, wherein β -carotene serves as the predominant constituent.

Recognizing the advantages associated with carotenoid compounds and the substantial carotenoid content within palm oil, the extraction of carotenoids from palm oil has garnered significant interest from the research community. The incorporation of carotenoid extraction methodologies through the esterification route is currently being refined, achieving a carotenoid purity level of 95 %. Methodological integrations such as solvolytic micellization (SM)-distillation-saponification adsorption [2] and the combination of saponification-solvent extraction-crystallization [3] are being explored. The technology for extracting carotenoids from palm can be synergistically aligned with the methyl ester (biodiesel) production process. This integrated technological approach is being developed as a strategy for diversifying palm-derived products, which may bolster food security initiatives. Moreover, the carotenoid concentrate obtained through the esterification route can enhance the value of biodiesel production.

This study undertook an investigation into the extraction of carotenoids from palm oil utilizing the liquid-liquid extraction technique (maceration) with an alcoholic solvent and deep eutectic solvent based on betaine as a co-solvent and explain the properties of DES that make them attractive for extraction processes, such as their biodegradability, low toxicity, and tunability. Additionally, an analysis was conducted to eliminate solvent impurity compounds present in carotene through washing and drying procedures.

2. Material dan methods

2.1. Material

The main raw material used in this research is Crude Palm Oil (CPO) obtained from Medan - Indonesia. The alcohols used were Methanol 99.80 %, Glycerol ≥ 99.97 %, Acetic acid > 99.97 %, NaOH 99.98 % and Betaine 99.95 % obtained from Sigma-Aldrich.

2.2. Methods

2.2.1. Transesterification reaction

The transesterification reaction of palm oil was conducted utilizing a

NaOH solution at a 1 % weight concentration, wherein 25 g of palm oil was reacted with methanol to generate a homogeneous sodium methoxide solution over the course of 1 h. Subsequently, sodium methoxide was introduced to the CPO: methanol mixture at a ratio of 1:6 within a three-neck flask, agitated at 250 rpm and maintained at a temperature of 60 °C for 1 h. The resultant methyl ester product was then separated from glycerol within a separatory funnel over a duration of 12 hours. The upper layer was identified as the methyl ester, while the lower layer was determined to be glycerol. The obtained methyl ester was subsequently washed with 500 ml of clean water, followed by a drying process at 105 °C for 30 minutes.

2.2.2. Preparation of Deep Eutectic Solvents (DES)

Deep Eutectic Solvent (DES) was synthesized by amalgamating Betaine with acetic acid and glycerol in a molar ratio of 1:1 to 1:6 (b/b) with 2 g Betaine: 2–12 g of acetic acid/glycerol subjected to thermal conditions of 70 °C for a duration of 30 minutes with a stirring rate of 300 rpm until a uniform and transparent solution was achieved [7]. The resultant clear DES liquid was subsequently characterized utilizing Fourier Transform Infrared Spectroscopy (FTIR) with a spectral range of 400–4000 cm^{-1} to elucidate the functional groups present within the compounds synthesized in the DES amalgamation. Additionally, Proton Nuclear Magnetic Resonance (^1H NMR) spectroscopy was employed to ascertain the composition of the compounds in the Deep Eutectic Solvents (DES) solution, while viscosity measurements were conducted using a Brookfield viscometer.

2.2.3. Extraction of carotenoids from methyl ester

An amount of 10 g of methyl ester derived from the transesterification technique was combined with 50 ml of methanol and Deep Eutectic Solvents (DES) in a 1:1 to 1:6 wt/weight ratio (Ester: DES). The obtained mixture was stirred with a magnetic stirrer for a time frame of 3 h at a rotational speed of 250 rpm under ambient thermal conditions. Consequently, the resultant product was permitted to undergo separation over a period of 24 hours, leading to the development of two discrete layers, wherein the superior layer consisted of the Ester and vitamin E constituents, while the inferior layer housed carotene. Furthermore, the carotene-rich lower layer of Deep Eutectic Solvents (DES) was subjected to re-extraction with a water: hexane mixture in a 1:1 vol/vol ratio for 15 minutes to eliminate any residual Deep Eutectic Solvents (DES) present in the carotene extract [8]. Following this, an evaporation process was conducted at temperatures ranging from 50 to 60 °C for 3 h to expunge the water content and any remaining Deep Eutectic Solvents (DES).

2.2.4. Analysis of methyl ester and total carotene

The assessment of methyl ester as a raw material was performed using Gas Chromatography-Mass Spectrometry (GC-MS) Shimadzu 2020, while the quantification of total carotenoid content was accomplished through a UV-Visible spectrophotometer. The analysis of the β -carotene compound was carried out utilizing High-Performance Liquid Chromatography (HPLC).

3. Results and discussions

3.1. Characterization of Crude Palm Oil and Palm Methyl Ester

In the current investigation, the characterization of Crude Palm Oil and Palm Methyl Ester was executed through Gas Chromatography to discern the constituents of Triglycerides, Diglycerides, Monoglycerides, and Esters present in both Crude Palm Oil and Palm Methyl Ester, while the carotene concentration was characterized using a UV-Visible Spectrophotometer. The findings obtained are illustrated in Table 1 below.

The data presented in Table 1 indicate that the carotene concentration in Crude Palm Oil (CPO) is quantified at 500 ppm, attributable to the binding of carotene compounds to triglycerides. Following the

Table 1

Characteristics of crude palm oil and palm methyl Ester.

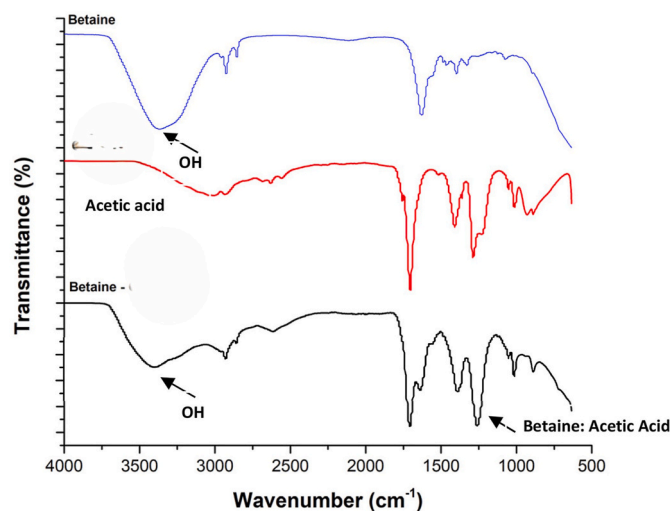
No	Parameter	Crude Palm Oil	Palm Methyl Ester
1	Triglycerides (%)	74.6516	0.502
2	Diglycerides (%)	8.0316	4.578
3	Monoglycerides (%)	0.0738	3.674
4	Ester (%)	17.243	91.245
5	Carotene (m g/ ml)	500	700

transesterification procedure utilizing a base catalyst, an elevation in carotene concentration was observed, as the carotene compounds initially bound within triglycerides were subsequently degraded into ester compounds.

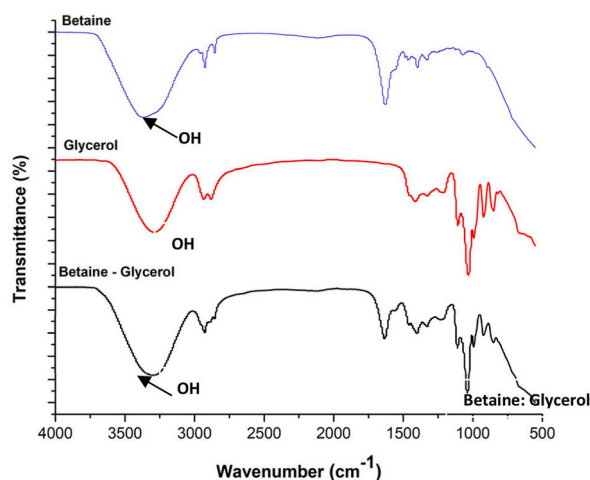
3.2. FTIR analysis

FTIR analysis of the DES compound was performed to discern the various functional groups within the compound. The analysis yielded spectral data represented graphically, as depicted in Fig. 1 below.

All the hydrogen bond donors used in these DES blends have



(a)



(b)

Fig. 1. FTIR spectra of DES betaine: acetic acid (a) and betaine: glycerol (b).

carboxylic acid groups in their structure at around the wave number 1750 cm^{-1} spectrum. On the other hand, in the FTIR spectrum of the hydrogen bond acceptor, betaine only one representative band corresponds to a hydroxyl group at around the wave number 3400 cm^{-1} . In addition, FTIR measurements were also used to confirm the formation of a new compound, via hydrogen bond formation, between betaine and the hydrogen bond donor. The FTIR spectra data of the eutectic blends showed intermolecular hydrogen bonding interactions between the hydrogen bond donor and betaine mostly in acetic acid and glycerol. It can be observed in Fig. 1 that for DES 1 eutectic blend, the carbonyl band of the HBD is initially located at low wave number values ($\sim 1700\text{ cm}^{-1}$) and expands and shifts to the highest values (1745 and 1748 cm^{-1}) [9]. In this DES blend there is a strong indication of new hydrogen bond formation, thus proving that new compounds are formed.

3.3. ^1H NMR analysis

Proton Nuclear Magnetic Resonance (^1H NMR) spectroscopy provides comprehensive structural insights regarding the hydrogen atoms within an organic molecule. It is noteworthy that not all ^1H nuclei experience identical spin inversion in response to radio frequency due to variations in their chemical or electronic environments, which engender chemical shifts. The ^1H NMR spectroscopic results for DES consisting of Betaine; acetic acid and DES formed from Betaine: glycerol in this study is exhibited in Fig. 2 below.

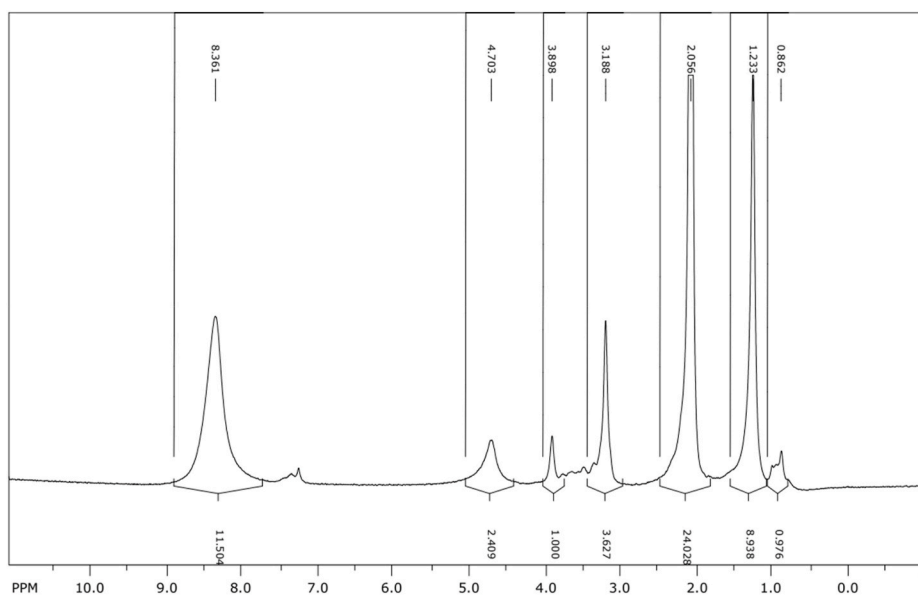
3.4. Physicochemical properties of DES

The effectiveness of compound extraction using Deep Eutectic Solvents (DES) is highly dependent on its dissolution properties. DES has the ability to give and receive protons and electrons, so it has the ability to form hydrogen bonds, which is closely related to its dissolution properties. Deep Eutectic Solvents (DES) has a high ability to perform hydrogen bonding interactions with the extracted compounds. Other DES properties that also have a great influence in extraction, are polarity and viscosity. Optimization of all these properties, makes Deep Eutectic Solvents (DES) a better solvent compared to other conventional solvents, such as water or ethanol [8].

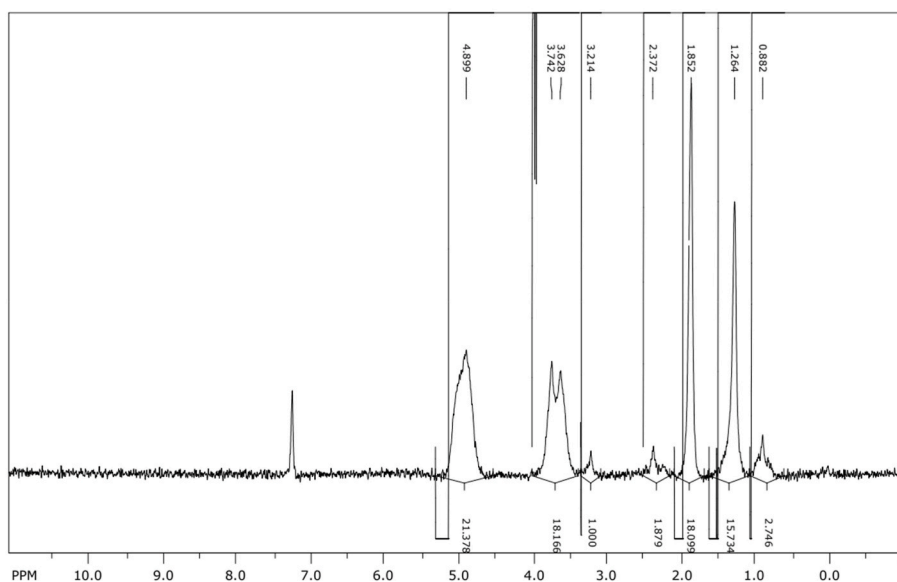
The viscosity of Deep Eutectic Solvents (DES) will influence the ability of Deep Eutectic Solvents (DES) to extract the desired target compound. The higher the viscosity value of DES, the more difficult it is for the target compounds to interact. The viscosity of DES is largely determined by hydrogen bonding, van der Waals, and electrostatic interactions. Kinetic energy from heating (rising temperature) can increase the frequency of intermolecular collisions resulting in weak intermolecular forces. The viscosity properties of NaDES can be adjusted by the addition of water. Water is a compound that can act as a giver and receiver of hydrogen bonds. With the addition of water, the hydrogen bond will also affect the NaDES molecule [10]. Dilution with water causes the HBD-HBA interaction to gradually weaken and even disappear. On the other hand, the addition of a small amount of water can reduce the viscosity of Deep Eutectic Solvents (DES) to water-like, thus increasing the conductivity up to 100 times [10].

Density values are necessary to understand the behavior of a liquid [11]. Density measurements are especially important in the field of fluid mechanics, mass transfer calculations and also for designing chemical processes [12]. Density is one of the important physical properties of ionic liquids in general and DES in particular [12].

According to Hayyan et al. (2014), the development of DES with low viscosity is desirable because Deep Eutectic Solvents (DES) has the potential to be an environmentally friendly medium [13]. The lower the viscosity value of Deep Eutectic Solvents (DES), the better Deep Eutectic Solvents (DES) is used as a solvent [14]. From Table 1 and it can be seen that the viscosity values of the two Deep Eutectic Solvents (DES) produced are 17.400 and 43.275 mPa s . This result shows that the viscosity of Deep Eutectic Solvents (DES) has a high value due to the mixing



(a)



(b)

Fig. 2. ^1H NMR DES betaine: Acetic acid and betaine: Glycerol.

process does not use water because with the addition of water, the hydrogen bonds will also affect the NaDES molecules. Dai et al. (2015) showed that dilution with water causes the HBD-HBA interaction to gradually weaken and even disappear. On the other hand, the addition of a small amount of water can reduce the viscosity of NaDES to water-like, thereby increasing the conductivity up to 100 times [10].

3.5. Effect of methyl ester ratio: DES on total carotene concentration

All types of carotenoids are hydrophobic, lipophilic compounds and are actually insoluble in water, but soluble in fat solvents. In animals and plants, carotenoids are crystal-like or amorphous solid and soluble in fat. The poor solubility of Deep Eutectic Solvents (DES) in water indicates

that Deep Eutectic Solvents (DES) in this study is a non-polar, or hydrophobic molecule. In contrast, methyl esters are non-polar organic substances. This information is important to create a biphasic system in liquid-liquid extraction [15]. In this study Deep Eutectic Solvents (DES) acts as a co-solvent by using the main solvent methanol which is semi-polar. The use of methanol in this study is because methanol is an organic compound so it is non-toxic, and does not pollute the environment. The carotene concentration results obtained are shown in Fig. 3 below.

Fig. 3 shows the effect of ester ratio: DES to the concentration of carotene obtained in the ester through extraction shows that the highest concentration at 2500 ppm in DES betaine: acetic acid compared to Deep Eutectic Solvents (DES) betaine: glycerol resulted in 2287 ppm. This is

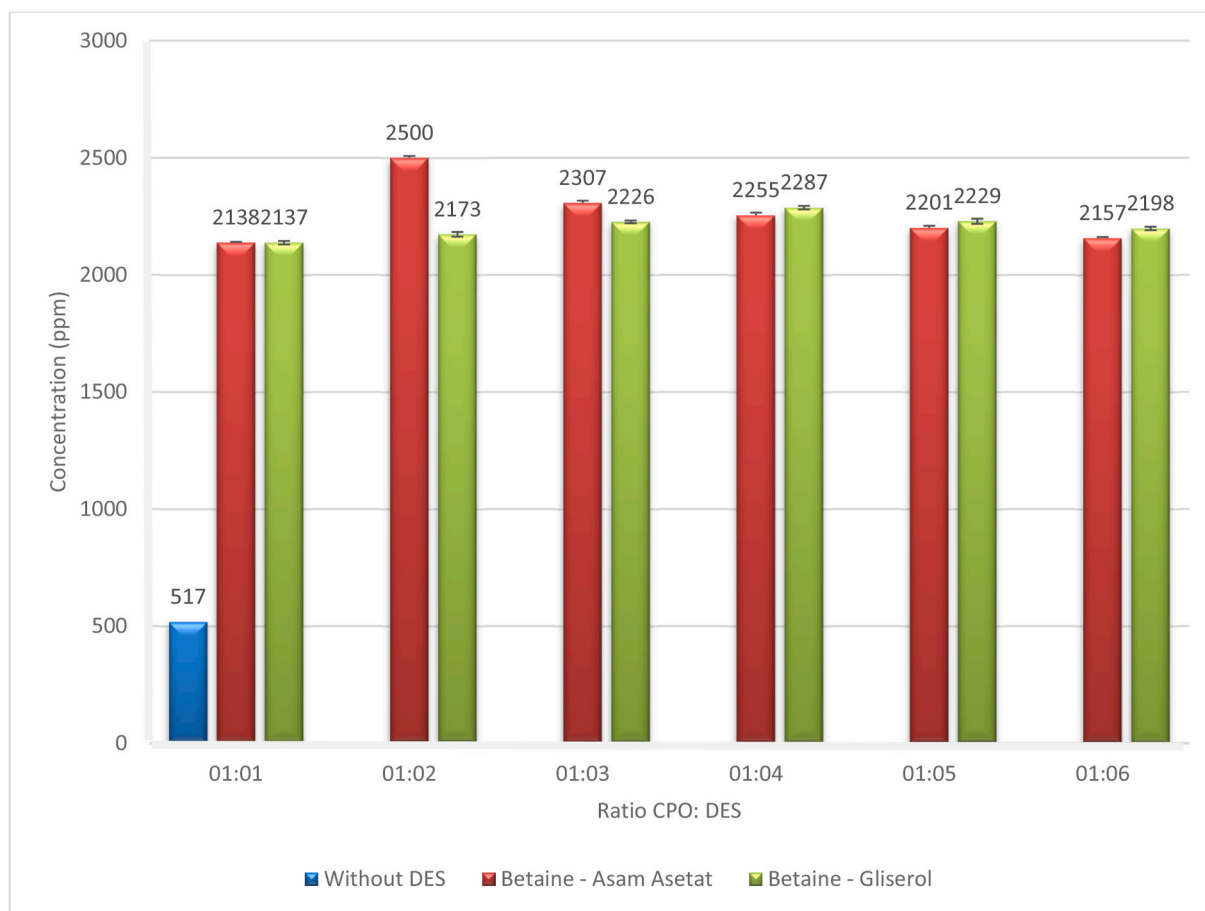


Fig. 3. Effect of ratio Ester: Deep Eutectic Solvents (DES) to carotene concentration.

because the value of density and viscosity in Deep Eutectic Solvents (DES) betaine: acetic acid is lower than Deep Eutectic Solvents (DES) betaine: glycerol it can be seen in Table 2 this is because the higher the viscosity value of NaDES, the more difficult the target compound will interact [16]. The viscosity of Deep Eutectic Solvents (DES) is largely determined by hydrogen bonds, van der Waals, and electrostatic interactions. In addition, the number of carboxylic bonds is small (-OH) in Deep Eutectic Solvents (DES) betaine: acetic acid while in Deep Eutectic Solvents (DES) betaine: glycerol (3-OH). In addition, an extraction treatment without Deep Eutectic Solvents (DES) (methanol only) was carried out resulting in a carotene concentration of 517 ppm. These results indicate that the addition of Deep Eutectic Solvents (DES) as a co-solvent can increase the total concentration of carotene product in the ester. In addition, Deep Eutectic Solvents (DES) polarity is an important factor in determining the separation efficiency which is influenced by the interaction between carotene solute and Deep Eutectic Solvents (DES) [11,12]. Here, the polarity of Deep Eutectic Solvents (DES) is hydrophobic and becomes the driving force for solute (carotene) transfer.

In addition, solvents with high density can affect the solubility of carotenoids. Higher density is usually associated with increased

solubility, which may inhibit the solvent's ability to solubilize nonpolar compounds such as carotenoids. As a result, the amount of carotenoids successfully extracted is less.

Meanwhile, when compared with research conducted by Aini and Manurung et al. (2020), carotenoids in palm oil were extracted using a solvent extraction method with methanol and a Deep Eutectic Solvent (DES) as a co-solvent. DES which used was DL-Menthol-based DES as a Hydrogen Bond Donor (HBD) and acetic acid and lauric acid as Hydrogen Bond Acceptors (HBA) which were reacted with 1: 1 and 2: 1 M ratio The optimum conditions for carotene extraction in this study were 212.7 ppm using DES MLA, while for the concentration of β -carotene obtained 0.156 ppm [17]. These results showed that the DES solvent as a co-solvent can increase the yield of carotene extraction in crude palm oil. When compared to this study, DES made from betaine and acetic acid has unique properties that make it effective in extraction. Betaine is known as an environmentally friendly and non-toxic solvent, while acetic acid serves as a hydrogen bond donor (HBD) that enhances the solvent's ability. In the study, betaine-based DES showed good extraction efficiency for bioactive compounds from palm biodiesel. The optimal molar ratio for this DES was 1:2, with a high ability to retain carotene content during the extraction process from 1:1 to 1:6.

3.6. Effect of Deep Eutectic Solvents (DES) on β -carotene concentration

β -carotene as one of the minor components of crude palm oil, is a precursor of vitamin A and functions as a pro-vitamin A. β -carotene has positive biological effects that are beneficial to the body, including to overcome blindness due to xerophthalmia, prevent the onset of cancer, coronary heart disease, prevent premature aging process, increase body immunity and can also act as an antioxidant [13]. The results of the

Table 2

Physiochemical properties of DES.

No	Physiochemical	Betaine: acetic acid	Betaine: Glycerol
1	DES (30°C)	Colourless liquid	Colourless liquid
2	pH	5.0	5.5
3	Density 30 °C (g/ml)	1.0858	1.1506
4	Viscosity 30°C (Cp)	17.400	43.275

effect of without Deep Eutectic Solvents (DES) and Deep Eutectic Solvents (DES) addition on β -carotene concentration can be seen in Fig. 4 below.

In this study, the extraction of β -carotene in palm esters using hydrophobic DES using a combination of betaine and carboxylic acid. The migration of β -carotene compound was influenced by gravity, agitation using shaker and also influenced by similar polarity between Deep Eutectic Solvents (DES) and β -carotene as the target compound. This polarity also causes hydrogen bonding between Deep Eutectic Solvents (DES) and β -carotene with this extraction system, it is expected that β -carotene compounds in palm esters will be extracted well. Fig. 4 shows that the highest concentration of β -carotene was found at 33.63 ppm in DES betaine: acetic acid. This indicates that the use of DES betaine: acetic acid can increase the concentration of β -carotene compared to no DES and the use of DES betaine: glycerol.

3.7. Antioxidant activity test results

Antioxidants are small substances produced by the body and are able to inhibit or prevent a substrate caused by free radicals entering the body, but the amount produced is still often insufficient. The antioxidant activity of carotene compounds can be seen in Table 3.

Table 3 shows the free radical activity of carotene extracts from methyl esters. The IC50 results showed betaine: glycerol was more potent than betaine: acetic acid at about 61.07 $\mu\text{g}/\text{ml}$ and betaine: acetic acid 33.52 $\mu\text{g}/\text{ml}$. Several studies reported that high levels of α -carotene and β -Carotene can increase the antioxidant activity of a product. However, the antioxidant activity of Palm Methyl Ester was stronger than that of crude palm oil. This is expected, due to the change of *trans*- β -carotene to *cis*- β -carotene isomer during the transesterification process.

Table 3

DPPH antioxidant activity of carotene.

No	DES	Antioxidant Activity (IC50 ($\mu\text{g}/\text{mL}$))
1	Without DES	2684.10 $\pm$ 51.71
2	Betaine: Acetic acid	2813.27 $\pm$ 33.52
3	Betaine: Glycerol	3417.13 $\pm$ 61.07

4. Conclusion

In summary, DES is a promising alternative solvent to extract and concentrate vitamin A from esters. The optimum total carotene concentration obtained in this study was 2500 ppm or equivalent to a concentration of 3.43 times using DES betaine: acetic acid which means it has better solubility properties than DES betaine: glycerol. This also applies to the β -carotene pickup concentration resulting in 33.63 ppm. This study confirms that the use of Deep Eutectic Solvent (DES) as a green solvent in carotenoid extraction is a sustainable and efficient solution compared to conventional petrochemical-based solvents. DES offers various advantages, including high biodegradability, low toxicity, and environmental compatibility, making it a greener alternative for the natural materials extraction industry. The results showed that betaine: acetic acid-based DES has high carotenoid extraction performance due to its optimum polarity and viscosity in extracting carotenoids in Palm Methyl Ester (PME) further strengthening the eco-friendly approach by replacing fossil-based organic solvents. In addition to improving extraction efficiency, the combination of DES and PME also supports the principle of circular economy, especially in the palm oil and biodiesel industry.

CRedit authorship contribution statement

Renita Manurung: Funding acquisition, Formal analysis, Data

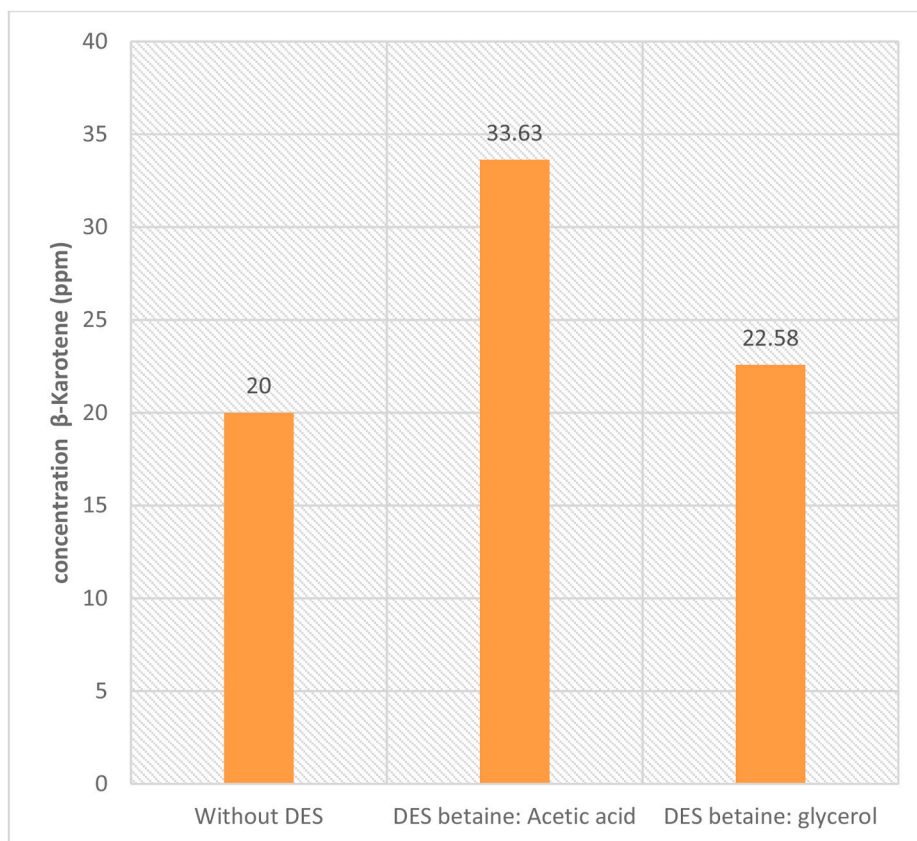


Fig. 4. The effect of DES on β -carotene concentration.

curation, Conceptualization. **Evi Naria:** Formal analysis, Data curation, Conceptualization. **Ilmi:** Formal analysis, Data curation, Conceptualization. **Alwi Gery Agustan Siregar:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

This statement is to certify that all Authors have seen and approved the manuscript being submitted revision. We warrant that the article is the Authors' original work. We warrant that the article has not received prior publication and is not under consideration for publication elsewhere. On behalf of all Co-Authors, the corresponding Author shall bear full responsibility for the submission. This research has not been submitted for publication nor has it been published in whole or in part elsewhere. We attest to the fact that all Authors listed on the title page have contributed significantly to the work, have read the manuscript, attest to the validity and legitimacy of the data and its interpretation, and agree to its submission to the Case Studies in Chemical and Environmental Engineering.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

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